B.TECH. CSE SUMMER INTERNSHIP • SESSION 2025–2026

Personal Document Q&A Assistant via Local RAG

Six-Week AI & NLP Training at InternCourse & End-to-End Pipeline Engineering

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AI Training Built a Local RAG Pipeline

Training Profile & Core Motivation

- **Host & Mentorship:** 6 weeks (15 Jun–18 Jul 2026) at InternCourse (ISO/AICTE) under MD Ankit Pandey and Mentor Pooja Parmar.
- **Hallucination & Privacy:** Static LLMs fabricate ungrounded answers on private docs and risk data exposure via third-party APIs.
- **Engineering Objective:** Build a local RAG assistant grounding answers strictly in uploaded PDF, DOCX, and TXT files.



SYSTEM ARCHITECTURE • END-TO-END RAG DATAFLOW

Seven-Stage Pipeline Grounds Generation in Local Vectors

1. Parse & Split

- **Clean Parsing:** PyMuPDF and pdfplumber extract clean text and tables.
- **Sliding Window:** Splits text into 500-token chunks with 50 overlap.

2. Embed & Index

- **Dense Vectors:** all-MiniLM-L6-v2 maps chunks into 384D semantic space.
- **HNSW Graph:** ChromaDB persists vectors for sub-15ms cosine lookup.

3. Search & QA

- **Cutoff Top-K:** Retrieves top K=4 chunks above a 0.65 cosine threshold.
- **Grounded LLM:** Local Llama-3-8B answers with strict page citations.

TECHNICAL STACK & ENVIRONMENT • ZERO CLOUD DEPENDENCY

Open-Source Stack Runs Entirely on Workstation Hardware

Core AI & NLP

- **LangChain:** Chains Python 3.11 loaders, splitters, and prompt rules.
- **MiniLM-L6-v2:** Encodes 384D semantic vectors in under 25ms per chunk.

DB & Local LLM

- **ChromaDB:** Serverless HNSW index storing vectors and page metadata.
- **Llama-3-8B:** Runs via Ollama on RTX 3060 (6GB VRAM) and 16GB RAM.

UI & DevOps

- **Streamlit UI:** Renders real-time PDF ingestion and Q&A citation views.
- **Docker & Git:** Containerized Linux build hosted on GitHub repository.

Source: Internship Report Ch. 5 (Section 5.1 & Table 5.1 Hardware/Software Specifications)

Modular Python Codebase Separates Config, Indexing, and QA

Repository Module	Core Class / Component	Key Parameters & Methods	Engineering Responsibility
config.py	RAGConfig (@dataclass)	CHUNK_SIZE=500, OVERLAP=50, TOP_K=4, THRESHOLD=0.65	Centralizes model hyperparameters, ChromaDB paths, and strict anti-hallucination system prompt.
ingestion.py	IngestionPipeline	load_document(), ingest_file(), RecursiveCharacterTextSplitter	Parses PDF/TXT files, splits text on hierarchical separators, attaches page metadata, and indexes into ChromaDB.
query_engine.py	RAGQueryEngine	retrieve(), generate_answer(), similarity_search_with_relevance_scores()	Computes cosine similarity, filters Top-K chunks, tracks retrieval latency (ms), and formats page citations.
app.py (UI Layer)	Streamlit Web Console	Drag-and-drop uploader, hyperparameter sliders, citation cards	Provides reactive dark/light dashboard with real-time vector count, chat history, and cosine score inspector.

AGILE WORK LOG • 15 JUNE 2026 – 18 JULY 2026

Six Weekly Sprints Delivered Parsing, Indexing, and UI

Weeks 1–2: Setup

- **Week 1 (Jun 15):** Set up Python 3.11, studied RAG papers, tested PyMuPDF.
- **Week 2 (Jun 22):** Built PDF/DOCX parser and tested text splitters.

Weeks 3–4: RAG

- **Week 3 (Jun 29):** Compared MiniLM vs BGE/mpnet; built ChromaDB schema.
- **Week 4 (Jul 06):** Tuned Top-K=4 search and anti-hallucination rules.

Weeks 5–6: QA

- **Week 5 (Jul 13):** Built Streamlit UI, chat memory, and latency profiler.
- **Week 6 (Jul 20):** Ran 50-query benchmark on 120-page test corpus.

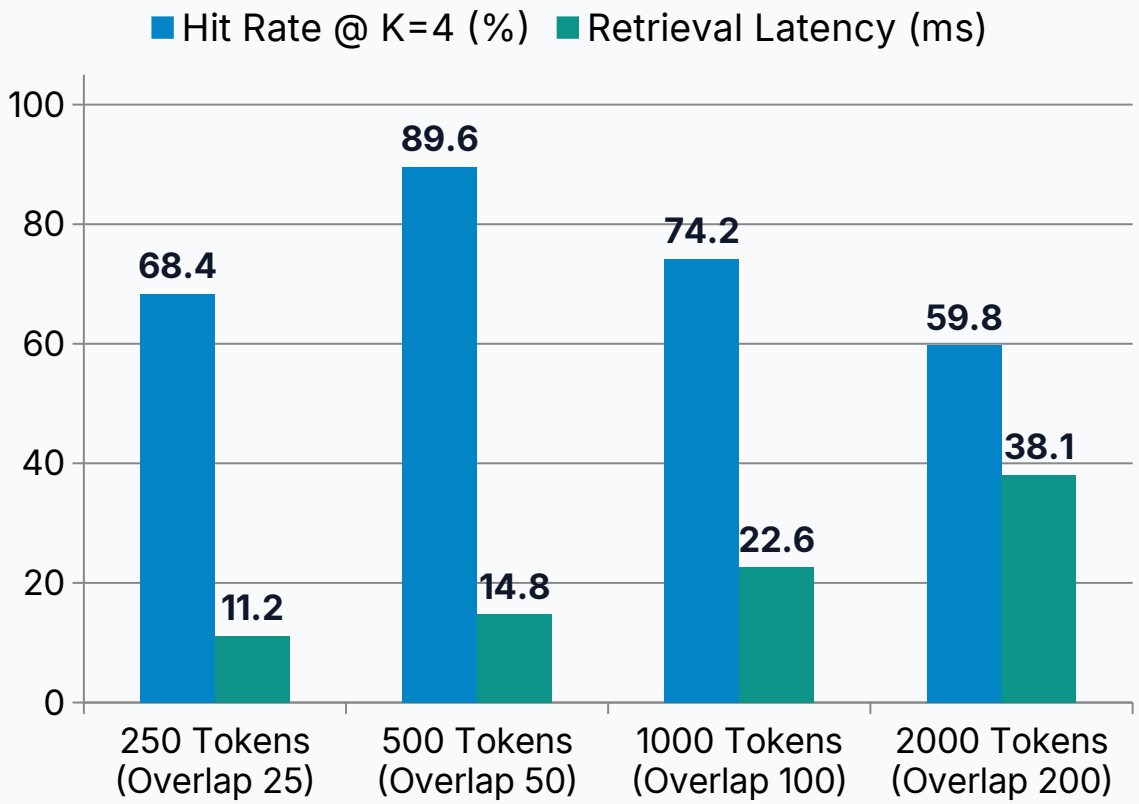
Source: Internship Report Ch. 6 (Table 6.1 Comprehensive 6-Week Internship Work Log) |

EMPIRICAL BENCHMARK • 120-PAGE CORPUS & 50 GROUND-TRUTH QUERIES

500-Token Chunks Achieved 89.6% Hit Rate at 14.8ms Latency

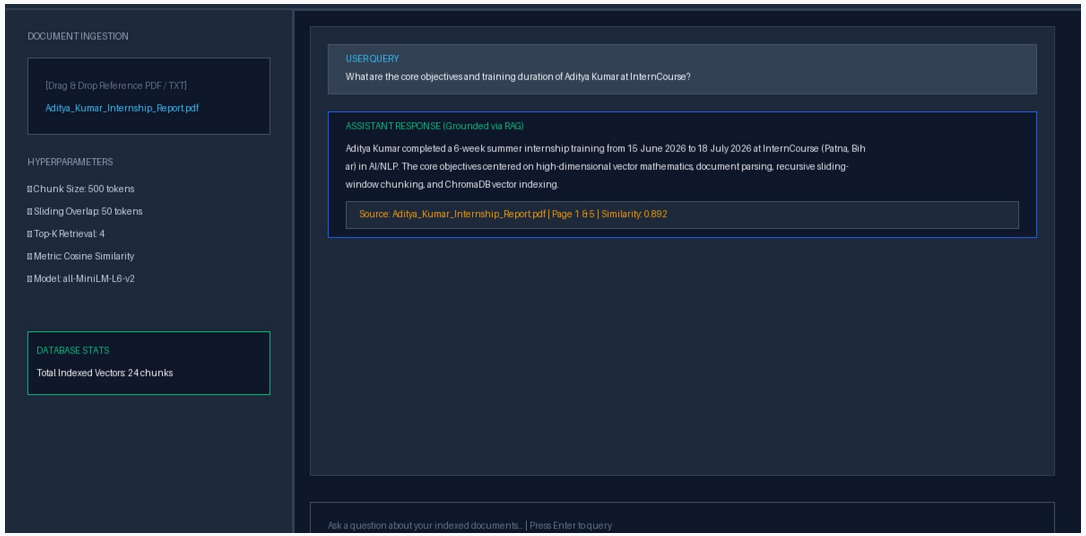
Chunk Size vs. Accuracy Findings

- **Optimal 500-Token Window:** 500 tokens (50 overlap) delivered peak 89.6% Hit Rate @ K=4 with fast 14.8ms retrieval latency.
- **Small Chunk Fragmentation:** 250-token chunks ran in 11.2ms but fell to 68.4% hit rate from context loss.
- **Large Chunk Dilution:** 1000–2000 token chunks diluted precision (74.2%, 59.8%) and raised latency.



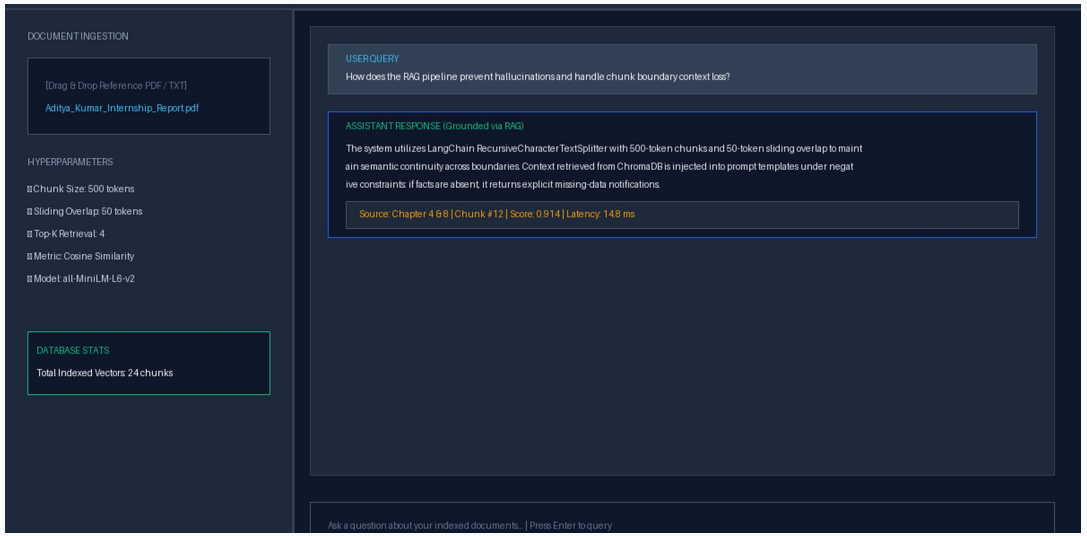
Source: Internship Report Ch. 6 (Section 6.2 & Table 6.2 Empirical Evaluation of Chunk Sizes) | Slide 7 of 10

Streamlit UI Displays Grounded Answers and Cosine Scores



Ingestion Console & Hyperparameter Controls

- **Live Indexing:** Drag-and-drop PDF uploader displays active ChromaDB chunks (24 vectors) and tunable sliders.



Real-Time Citation & Relevance Inspector

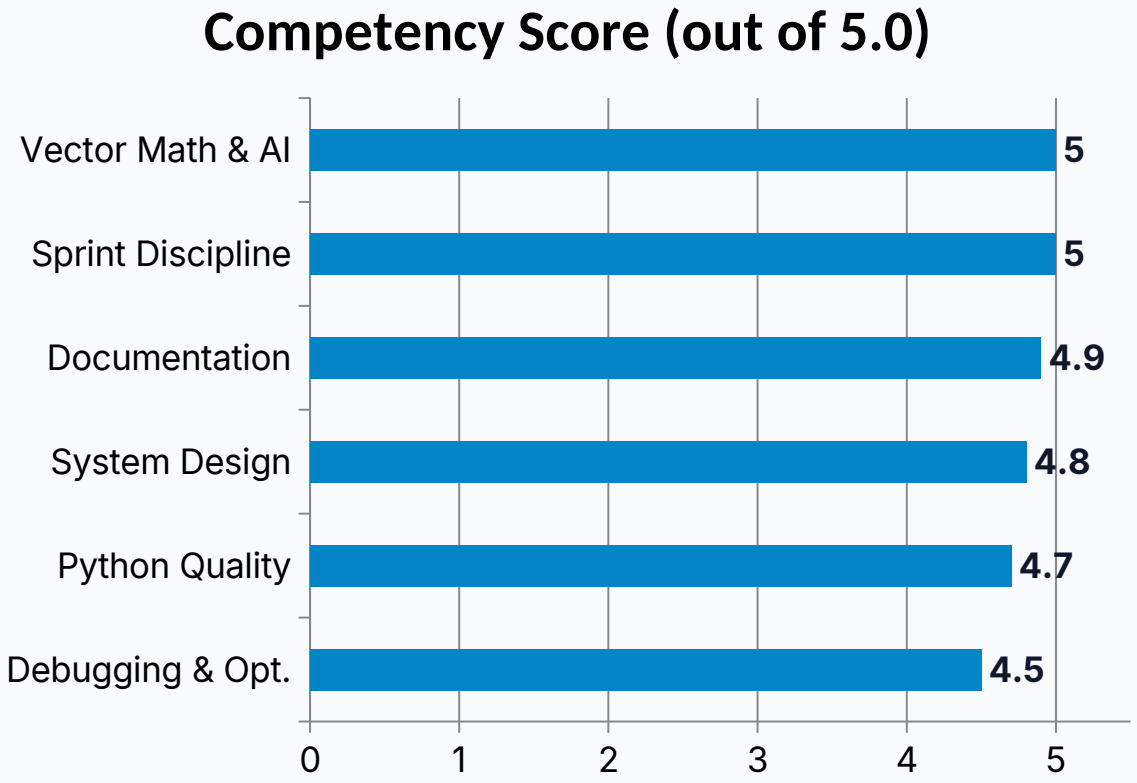
- **Verified Provenance:** Renders exact document page citations, cosine similarity score (0.914), and 14.8ms lookup latency.

Source: Internship Report Appendix C (Figures 11.2 & 11.3 Interactive Streamlit UI) | Slide 8 of 10

Recursive Splitting and Cutoffs Solved Context Fragmentation

Engineering Solutions & Future Roadmap

- **Boundary Continuity:** Used recursive separators and 50-token overlap to preserve full sentences.
- **Noise & Table Handling:** Applied a 0.65 cosine cutoff (Top-K=4) and parsed PDF tables to Markdown.
- **Future Enhancements:** Add BM25 + dense hybrid RRF search, BGE cross-encoders, and Tesseract OCR.



THANK YOU • QUESTIONS & DISCUSSION

Grounded AI Starts With Verifiable Local Retrieval.

Explore the full source code, Streamlit UI, and evaluation suite on GitHub.

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